John D. Soltis

https://orcid.org/0000-0002-0104-3593 | https://github.com/johnsoltis
https://www.linkedin.com/in/john-soltis/

RESEARCH POSITIONS:

Science & Technology Policy Fellow 2026 - 2027 Washington, DC
Legislative Branch, Cohort in AI
Host: U.S. House of Representatives
Sponsor: American Association for the Advancement of Science (AAAS)

Lasker Data Science Fellow 2025 - 2026 Baltimore, MD
Postdoctoral Prize Fellow
Space Telescope Science Institute

Pre-Doctoral Fellow 2023 - 2024 New York, NY
Advisor: Lehman Garrison
Center for Computational Astrophysics, Flatiron Institute

PhD Student 2019 - 2025 Baltimore, MD
Advisor: Michelle Ntampaka
Johns Hopkins University

Affiliate 2018 - 2019 Berkeley, CA
Advisor: Carlton Pennypacker
Lawrence Berkeley National Laboratory

Research Assistant 2015 - 2018 Ann Arbor, MI
Advisor: Dragan Huterer
University of Michigan

Summer Scholar 2015 Livermore, CA
Advisors: Jason Chou & John Heebner
Lawrence Livermore National Laboratory

EDUCATION:

Johns Hopkins University 2019 - 2025 Baltimore, MD
Krieger School of Arts & Sciences
Ph.D in Astronomy and Astrophysics

Johns Hopkins University 2019 - 2023 Baltimore, MD
Krieger School of Arts & Sciences
M.A. in Physics

University of Michigan 2014 - 2018 Ann Arbor, MI
Honors Program, College of Literature, Science, and the Arts
B.S. in Physics and Mathematical Physics

SKILLS & INTERESTS:

General
Machine Learning, Uncertainty Quantification, AI Interpretability,
Hierarchical Bayesian Modeling, Data Analysis, Cosmology, Milky Way Astronomy

Software Packages & Languages
Python, Pytorch, Tensorflow, Unix, LaTeX, MATLAB, Microsoft Excel

FIRST AUTHOR PUBLICATIONS:

1. **J. Soltis**, "Learning Gaia: A Generative Model Trained on the Gaia Data Release 3 Source Catalog", *arXiv e-prints*, arXiv:2609.38326, (2026)

2. **J. Soltis**, M. Ntampaka, B. Diemer, J. ZuHone, S. Bose, A. M. Delgado, B. Hadzhiyska, C. Hernández-Aguayo, D. Nagai, H. Trac. "A Multi-Wavelength Technique for Estimating Galaxy Cluster Mass Accretion Rates", *The Astrophysical Journal*, 985, 2, 212, 12 pp. (2025)

3. **J. Soltis**, L. Garrison. "Self-Similar Mass Accretion Histories in Scale-Free Simulations", *Monthly Notices of the Royal Astronomical Society*, 532, 2, 1729-1743, (2024)

4. **J. Soltis**, M. Ntampaka, J. Wu, J. ZuHone, A. Evrard, A. Farahi, M. Ho, D. Nagai. "A Machine Learning Approach to Enhancing eROSITA Observations", *The Astrophysical Journal*, 940, 1, 60, 17 pp. (2022)

5. **J. Soltis**, S. Casertano, A. G. Riess. "The Parallax of Omega Centauri Measured from Gaia EDR3 and a Direct, Geometric Calibration of the Tip of the Red Giant Branch and the Hubble Constant", *The Astrophysical Journal Letters*, 908, L5 (2021)

6. **J. Soltis**, A. Farahi, D. Huterer, C. M. Liberato. "Percent-Level Test of Isotropic Expansion Using Type Ia Supernovae", *Phys. Rev. Lett.*, 122, 091301 (2019)

CO-AUTHOR PUBLICATIONS:

1. B. E. M. Davis, **J. Soltis**. "On American scientific isolationism: A conversation with Dr. Sudip Parikh, CEO of the American Association for the Advancement of Science", *MIT Science Policy Review*, Vol. 7, pg. 158-167. (2026)

2. M. Ntampaka, S. Burke-Spolaor, I. Ciucă, A. M. Delgado, K. G. Iyer, K. Lockhart, A. Plunkett, M. López-Morales, A. A. Mahabal, S. E. Mullally, R. Raj, J. Reerink, J. Smith, J. S. Speagle, **J. Soltis**, J. F. Wu, M. Yunus."How to Craft the Right Language AI Policy For Your Research Group (Some Assembly Required)", *arXiv e-prints*, arXiv:2607.20836, (2026)

3. A. M. Delgado, M. Ntampaka, S. Bose, F. Ferlito, B. Hadzhiyska, L. Hernquist, **J. Soltis**, J. F. Wu, M. Yunus, J. ZuHone. "Estimating the Triaxiality of Massive Clusters from 2D Observables in MillenniumTNG with Machine Learning", *The Astrophysical Journal*, 1002, 2, 142, 23 pp. (2026)

4. M. Ntampaka, A. Ciprijanovic, A. M. Delgado, **J. Soltis**, J. F. Wu, M. Yunus, J. ZuHone. "The Importance of Being Adaptable: An Exploration of the Power and Limitations of Domain Adaptation for Simulation-Based Inference with Galaxy Clusters", *arXiv e-prints*, arXiv:2510.09748, (2025)

5. B. E. M. Davis, M. Razavi-Mohseni, **J. Soltis**, H. N. Zhang, E. Kavanagh. "International STEM Graduate Students: A Key to Strengthening the American Economy and Building Competitiveness", *Journal of Science Policy and Governance*, 25, 1, (2024)

6. M. Ho, **J. Soltis**, A. Farahi, D. Nagai, A. Evrard, M. Ntampaka. "Benchmarks and Explanations for Deep Learning Estimates of X-ray Galaxy Cluster Masses", *Monthly Notices of the Royal Astronomical Society*, 524, 3, 3289-3302, (2023)

7. W.C. Wan, G. Malamud, A. Shimony, C.A. Di Stefano, M.R. Trantham, S.R. Klein, **J.D. Soltis**, D. Shvarts, R.P. Drake, C.C. Kuranz. "Impact of ablator thickness and laser drive duration on a platform for supersonic, shockwave-driven hydrodynamic instability experiments", *High Energy Density Physics*, 22, (2017)

SELECTED TALKS AND POSTERS:

1. *Learning Gaia: A Flows-based Model Trained on the Gaia Data Release 3 Source Catalog.* 2026 IAIFI Summer Workshop, NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI). August 12th, 2026.
2. A Multi-Wavelength Technique for Estimating Galaxy Cluster Mass Accretion Rates. CAMELS + Baryon Pasting Meeting @ Osaka, Osaka University. April 9th, 2026.
3. **J. Soltis**, M. Ntampaka. "Direct Estimation of Galaxy Cluster Mass Accretion Rates using Machine Learning", American Astronomical Society Meeting #245, id. 412.02D. *Bulletin of the American Astronomical Society*, Vol. 57, No. 2 e-id 2025n2i412p02 (2025)
4. *Direct Estimation of Galaxy Cluster Mass Accretion Rates using Machine Learning.* Cosmology and Galaxy Astrophysics with Simulations and Machine Learning 2024 Conference, Flatiron Institute. December 9th, 2024.
5. *Direct Estimation of Galaxy Cluster Mass Accretion Rates using Machine Learning.* Galaxy Cluster Group Meeting, Center for Astrophysics, Harvard-Smithsonian. October 22nd, 2024.
6. *Direct Estimation of Galaxy Cluster Mass Accretion Rates using Machine Learning.* Yale Data Science X Astronomy - Astrophysics Seminar, Yale University. September 17th, 2024.
7. *Estimating Galaxy Cluster Mass Accretion Rates from Observations using Machine Learning.* 2024 AstroAI Workshop, Center for Astrophysics, Harvard-Smithsonian. June 17-21, 2024.
8. *Galaxy Cluster Dynamical State, Follow-Up Observations, and Machine Learning.* Machine Learning Seminar, Argelander Institute for Astronomy, Bonn. June 6th, 2024.
9. *Galaxy Cluster Mass Accretion History.* 6th Neighborhood Workshop, Pennsylvania State University. April 25th, 2024.
10. *Testing the Robustness of Mass Accretion Histories in Scale-Free Simulations.* Merging Clusters Workshop, Yonsei University. December 21st, 2023.
11. **J. Soltis**, M. Ntampaka. "Predicting Follow-Up Observations of Galaxy Clusters Using Machine Learning", American Astronomical Society Meeting #240, id. 139.19. *Bulletin of the American Astronomical Society*, Vol. 54, No. 6 e-id 2022n6i139p19 (2022)

RESEARCH PROJECTS:

AI Applications in Milky Way Astronomy 2025 - 2026
Space Telescope Science Institute
Trained generative AI foundation model on *Gaia* satellite observations of more than 1 billion Milky Way and nearby stars.

Deep Learning Applications in Galaxy Cluster Cosmology 2021 - 2026
Johns Hopkins University & Space Telescope Science Institute *PhD Advisor: Michelle Ntampaka*
Using convolutional neural networks, simulation-based inference, and machine learning interpretability methods to characterize galaxy cluster properties and improve observations.

Robustness of Cosmological Simulations 2023 - 2024
Flatiron Institute *Advisor: Lehman Garrison*
Investigated the robustness of dark matter halo mass accretion rates in cosmological simulations.

Tip of Red Giant Branch Calibration 2019 - 2020
Johns Hopkins University & Space Telescope Science Institute *Advisor: Adam Riess*

Measured the Hubble constant using updated position data of Milky Way stars.

Machine Learning Applications in Wildfire Detection 2018 - 2019
Lawrence Berkeley National Laboratory *Advisor: Carl Pennypacker*
Used images from wildfire detection cameras in California and Nevada to train a convolutional neural network to detect wildfires early.

Testing Statistical Isotropy with Type Ia Supernovae 2017 - 2018
University of Michigan *Advisor: Dragan Huterer*
Implemented a novel and robust test of statistical isotropy in the Universe using type Ia supernovae residuals.

Simulation of Laser-Driven Plasma Instabilities 2015 - 2016
University of Michigan *Advisor: Matthew Trantham*
Simulated laser-driven plasma instabilities with a variety of experimental conditions. Results were used to improve experimental design.

National Ignition Facility Support 2015
Lawrence Livermore National Laboratory *Advisors: John Heebner & Jason Chou*
Improved the accuracy of laser waveform generation in order to aid experiments at the National Ignition Facility.

EXTRACURRICULARS:

MIT Science Policy Review 2025-Present
Served as an interviewer for Volume 7, publishing a discussion on scientific isolationism with the CEO of the AAAS. Currently serving as an Executive Editor.

Institute of Electrical and Electronics Engineers AI Policy Committee 2025 - 2026
Helped formulate responses to Requests for Information from Senators and the Department of Energy.

Science Policy and Diplomacy Group Congressional Visit Day 2026
Helped organize and train groups for Congressional Visit Day. On the Hill, advocated for the Protecting Free Vaccines Act of 2025, which would preserve vaccine coverage and access.

Science Policy and Diplomacy Group Maryland Assembly Visit Day 2026
Helped organize and train the groups for meeting with members of the Maryland Assembly. Advocated for oversight of the Human-Relevant Research Fund and MD Law Chapter 448.

Advocacy Training Seminar 2025
Conducted an informational session for graduate students at Johns Hopkins related to Federal science policy and the role of scientists in advocacy.

Science Policy and Diplomacy Group Congressional Visit Day 2025
Advocated for robust science funding and against severe budget cuts

Science Policy and Consulting Career Panel 2025
Hosted five person panel on science policy careers for Johns Hopkins graduate students

Institute of Electrical and Electronics Engineers Congressional Visit Day 2025
Advocated for science funding, small business funding, and improved STEM workforce policies

American Astronomical Society Congressional Visit Day 2024
Met with Members of Congress and their staffs to advocate for increased NASA and NSF funding

Institute of Electrical and Electronics Engineers Congressional Visit Day 2024
Advocated for NSF funding and improved STEM workforce policies

Science Policy and Diplomacy Group Congressional Visit Day 2024
Advocated for the Keep STEM Talent Act of 2023

Graduate Representative Organization 2021 - 2023
General Council Representative for the Physics & Astronomy Department

Maryland Space Grant Observatory Fellow 2021 - 2022
Hosted open house events, trained perspective observers on the telescope, and helped run the MDSGC symposium.

Public Forum Debate Judge 2020 - 2022
Judged for Michigan Interscholastic Forensics Association Public Forum League

Michigan Journal of International Affairs 2014 - 2016
Writer for the Asia Region